

Project Proposal

Emergency departments across the United States face a chronic crisis of overcrowding, undertriage, and dangerously long wait times that directly impact patient outcomes. The core problem is not always a shortage of staff but often a shortage of decision support. When a patient walks into an ER, nurses and physicians must rapidly assess acuity (severity of condition) with incomplete information, high cognitive load, and time pressure. Mistakes in triage can result in high-acuity patients waiting too long, or low-acuity patients consuming critical resources.

TriageIQ is a machine learning system that predicts patient wait time buckets and acuity levels at the point of intake, using structured clinical data (chief complaint, vital signs, demographics, time of arrival, day of week) combined with free-text symptom descriptions. Critically, the system is not a black box as it uses SHAP (SHapley Additive exPlanations) values to surface the top features driving each prediction, so clinicians can interrogate and trust the model's reasoning rather than blindly follow it.

The project is built on the NHAMCS (National Hospital Ambulatory Medical Care Survey), a nationally representative public dataset from the CDC containing over 300,000 ER visit records with rich intake and outcome variables. This eliminates data access barriers and grounds the work in real-world clinical diversity.

The technical pipeline includes: (1) feature engineering from structured intake fields (by July 1), (2) TF-IDF or biomedical embedding of chief complaint text (by July 5), (3) an XGBoost ensemble model for acuity/wait prediction (by July 12), (4) SHAP explainability layer (by July 15), and (5) a Streamlit web interface simulating an intake screen (by July 25). The addition of NLP on chief complaint text, and demonstrating it improves over structured features alone, is the novel technical contribution beyond a standard classification task.

This project is especially relevant to UCSD, which operates UC San Diego Health, one of the busiest Level I trauma centers in San Diego County. The social impact here is immediate and legible as it optimises triage tools to save lives.